Weird Logic

June 30, 2025

1 Examples

Correctness of Doppler in hyper triple:

$$\{\forall x \in Var, x(\mathcal{P}) = x(L)\} \left[\begin{array}{l} p \in \mathcal{P} : C(p) \\ l \in L : l \end{array} \right] \left\{ \begin{array}{l} \forall\, l \in L, \exists\, p \in \mathcal{P} \\ x(p) = x(l) \end{array} \right\}$$

For any program `l` written in the language `L`, there always exists a payload `p` such that the program `C` and `l` execute the same behaviour.

1.1 `if-else`

Given a Doppler attack language L:

$$\begin{array}{rcl} L & \rightarrow & S_1 : x = x + 1 \\ & & \mid S_2 : x = x + 2 \end{array}$$

and vulnerable program C:

```
choose a
if a then
    x = x+1
else
    x = x+2
```

write the triple:

$$\{x(\mathcal{P}) = x(L)\} \left[\begin{array}{lll} p \in \mathcal{P} & , & C(p) \\ l \in L & , & l \end{array} \right] \left\{ \begin{array}{c} \forall l \in L, \exists p \in \mathcal{P} \\ x(p) = x(l) \end{array} \right\}$$

where $P = \{a_\top, a_\bot\}$,

$$\{x(\mathcal{P}) = x(L)\} \left[\begin{array}{lll} p \in \{a_\top, a_\bot\} & , & C(p) \\ l \in S_1 \mid S_2 & , & l \end{array} \right] \left\{ \begin{array}{c} \forall l \in L, \exists p \in \mathcal{P} \\ x(p) = x(l) \end{array} \right\}$$

Apply GrmDisj-rule:

$$\frac{\begin{array}{c} L \to L_1 \mid L_2, L \notin L_1, L \notin L_2 \\ \{P\}\ [C(\mathcal{P}); L_1]\ \{Q\} \quad \{P\}\ [C(\mathcal{P}); L_2]\ \{Q\} \end{array}}{\{P\}\ [C(\mathcal{P}); L]\ \{Q\}}$$

Partition S_1 and S_2:

$$\forall i \in \{1,2\}, \{x(\mathcal{P}) = x(L)\} \left[\begin{array}{lll} p \in \{a_\top, a_\bot\} & , & C(p) \\ l \in S_i & , & l \end{array} \right] \left\{ \begin{array}{c} \forall l \in S_i, \exists p \in \mathcal{P} \\ x(p) = x(l) \end{array} \right\}$$

Apply Weaken-rule:

$$\frac{\mathcal{P} \supseteq \mathcal{P}_1 \quad \{P\}\ [C(\mathcal{P}_1); L]\ \{Q\}}{\{P\}\ [C(\mathcal{P}); L]\ \{Q\}}$$

Let $\mathcal{P}_1 = \{a_\top\}; \mathcal{P}_2 = \{a_\bot\}$, shrink $\mathcal{P}$:

$$\forall i \in \{1,2\}, \{x(\mathcal{P}) = x(L)\} \left[\begin{array}{lll} p \in \mathcal{P}_i & , & C(p) \\ l \in S_i & , & l \end{array} \right] \left\{ \begin{array}{c} \forall l \in S_i, \exists p \in \mathcal{P}_i \\ x(p) = x(l) \end{array} \right\}$$

Apply lang-rule:

$$\frac{\exists l, L = \{l\} \quad \{P\}\ [C(\mathcal{P}); l]\ \{\exists p \in \mathcal{P}, Q(l,p)\}}{\{P\}\ [C(\mathcal{P}); L]\ \{\forall l \in L, \exists p \in \mathcal{P}, Q(l,p)\}}$$

Apply payload-rule:

$$\frac{\exists p \in \mathcal{P},\ \{P\}\ [C(p); l]\ \{Q(l,p)\}}{\{P\}\ [C(\mathcal{P}); l]\ \{\exists p \in \mathcal{P}, Q(l,p)\}}$$

1.2 `while-choose`

Given a Doppler attack language L:

$$\begin{array}{rcl} L & \rightarrow & S_1 \\ S_1 & \rightarrow & x = x + 1; S_1 \\ & & \mid \epsilon \end{array}$$

or vulnerable program C:

```
while (choose b) do
    x = x+1
```

write the triple for C:

$$\{\forall l, p, x(p) = x(l)\} \left[\begin{array}{lll} p \in \mathcal{P} & , & C(p) \\ l \in L & , & l \end{array} \right] \left\{ \begin{array}{c} \forall l \in L, \exists p \in \mathcal{P} \\ x(p) = x(l) \end{array} \right\}$$

where $\mathcal{P} := \mathbb{B}^*$
Apply weaken-rule and shrink $\mathcal{P}$ to $\{b_\top *; b_\bot\}$
Apply LeftReg-rule:

$$\frac{\begin{array}{c} \mathcal{P} = \{p_\top *; p_\bot\} \\ \{P \wedge \mathcal{P}_1 = \{p_\top\}\}\ [C'(\mathcal{P}_1); L_1]\ \{P\} \end{array}}{\{P\} \left[\begin{array}{l} \texttt{while}\ \mathcal{P}\ \texttt{do}\ C' \\ L \rightarrow (L_1; L) \mid \epsilon \end{array} \right] \{P \wedge \exists p', p = p'; p_\bot\}} \text{LeftReg}$$

Seq-rule (should have more constraints):

$$\frac{\{P\}\,[C_1(\mathcal{P}_1),\ L_1]\,\{H\} \quad \{H\}\,[C_2(\mathcal{P}_2),\ L_2]\,\{Q\}}{\{P\} \left[\begin{array}{lll} p \in \mathcal{P}_1 \times \mathcal{P}_2 & , C_1(p_1); C_2(p_2) \\ l \in L_1; L_2 & , l \end{array} \right] \{Q\}} Seq$$

SeqLeft-rule:

$$\frac{\begin{array}{c} \{P\}\ [C_1(\mathcal{P}_1); L \rightarrow (L_1; L) \mid \epsilon]\ \{H\} \\ \{H\}\ [C_2(\mathcal{P}_2); L \rightarrow L_2]\ \{Q\} \end{array}}{\{P\} \left[\begin{array}{lll} p \in \mathcal{P}_1 \times \mathcal{P}_2 & , C_1(p_1); C_2(p_2) \\ l \in L \rightarrow (L_1; L) \mid L_2 & , l \end{array} \right] \{Q\}} \text{SeqLeft}$$

SeqRight-rule:

$$\frac{\begin{array}{c}\{P\}\ [C_1(\mathcal{P}_1); L \to L_2]\ \{H\} \\ \{H\}\ [C_2(\mathcal{P}_2); L \to (L_1; L) \mid \epsilon]\ \{Q\}\end{array}}{\{P\}\ \left[\begin{array}{ll} p \in \mathcal{P}_1 \times \mathcal{P}_2 & , C_1(p_1); C_2(p_2) \\ l \in L \to (L; L_1) \mid L_2 & , l \end{array}\right]\ \{Q\}}\text{SeqRight}$$

Apply RightReg-rule:
$Q' : \forall l \in L, p \in \mathcal{P}, x(p) = x(l) \wedge \exists p', p = p'; p_\perp$
$\implies Q : \forall l \in L, \exists p \in \mathcal{P}, x(p) = x(l)$
Get:

$$\{\forall l, p, x(p) = x(l) \wedge \mathcal{P} = \{p_\top\}\} \left[\begin{array}{ll} p \in \mathcal{P} & ,\ x = x + 1 \\ l \in \{x = x + 1\} & ,\ l \end{array}\right] \left\{\begin{array}{c} \forall l \in L, p \in \mathcal{P} \\ x(p) = x(l) \end{array}\right\}$$

Apply lang-rule and payload-rule.

1.3 `do-while`

Given a Doppler attack language L:

$$\begin{array}{rcl} L & \to & S_1 \\ S_1 & \to & x = x + 1 \\ & & \mid S_1; S_1 \end{array}$$

or language L':

$$\begin{array}{rcl} L & \to & x = x + 1; S_1 \\ S_1 & \to & x = x + 1; S_1 \\ & & \mid \epsilon \end{array}$$

and vulnerable program C:

```
do
    choose a
    x = x+1
while a
```

or vulnerable program C':

```
b=1
while b>0 do
```

```
    b--
    choose b
    x = x+1
```

or vulnerable program C'':

```
c=true
while c do
    choose c
    x = x+1
```

TODO: The mapping between program and language is not isomorphic. One solution is to restrict the language to left-regular format when it can be converted. For example, restrict the language to L' as shown above.

Write the triple for C:

$$\{x(\mathcal{P}) = x(L)\} \left[\begin{array}{ll} p \in \{\epsilon\} \times \mathcal{P}' & , \ C_1(\epsilon); C_2(p') \\ l \in \{L \to x = x + 1; S_1\} & , \ l \end{array} \right] \left\{ \begin{array}{c} \forall l \in L, \exists p \in \mathcal{P} \\ x(p) = x(l) \end{array} \right\}$$

where $\mathcal{P} := \mathbb{B}*$, $\mathcal{P} := \{\epsilon\} \times \mathcal{P}'$, and $p' \in \mathcal{P}'$;

$C_1 := x = x + 1$;

$C_2 :=$ `while` *choose a* `do` $x = x + 1$

Apply Seq-rule:

$$(1)\, \{x(\mathcal{P}) = x(L_1)\} \left[\begin{array}{ll} p \in \{\epsilon\} & , \ C_1(p) \\ l \in \{L_1 \to x = x + 1\} & , \ l \end{array} \right] \left\{ \begin{array}{c} \forall l \in L_1, \exists p \in \mathcal{P} \\ x(p) = x(l) \end{array} \right\}$$

$$(2)\, \{x(\mathcal{P}') = x(S_1)\} \left[\begin{array}{ll} p \in \mathcal{P}' & , C_2(\mathcal{P}') \\ l \in \{S_1 \to x = x + 1; S_1 | \epsilon\} & , \ l \end{array} \right] \left\{ \begin{array}{c} \forall l \in S_1, \exists p \in \mathcal{P}' \\ x(p) = x(l) \end{array} \right\}$$

(2) is the same as the `while-choose` example.

1.4 `choose-while`

Given a Doppler attack language L:

$$\begin{array}{rcl} L & \to & S_1 \\ S_1 & \to & x = x + 1; S_1 \\ & & | \ \epsilon \end{array}$$

or L':

$$\begin{array}{rcl} L & \rightarrow & \texttt{for}\ i\ \texttt{in}\ b\ \texttt{do}\ S_1 \\ S_1 & \rightarrow & x = x + 1 \end{array}$$

and vulnerable program C:

```
choose b
while b>0 do
    b--
    x = x+1
```

or vulnerable program C':

```
choose b
while b>0 do
    b = b-2;
    x = x+1
```

Write the triple for C:

$$\{x(\mathcal{P}) = x(L)\} \left[\begin{array}{lcl} p \in \mathcal{P} & , & C(p) \\ l \in L & , & l \end{array} \right] \left\{ \begin{array}{c} \forall l \in L, \exists p \in \mathcal{P} \\ x(p) = x(l) \end{array} \right\}$$

where $\mathcal{P} := \{b \mid b \in \mathbb{N}\}$.
Borrow the HP rule from unrealisability logic:

$$\frac{\begin{array}{c} \{P\}\,[C(\mathcal{P}); L]\,\{Q\} \vdash \quad \{P\}\,[C(\mathcal{P}); S_1]\,\{Q\} \\ \ldots \\ \{P\}\,[C(\mathcal{P}); L]\,\{Q\} \vdash \quad \{P\}\,[C(\mathcal{P}); S_n]\,\{Q\} \quad L \rightarrow S_1|...|S_n \end{array}}{\vdash \{P\}\,[C(\mathcal{P}); L]\,\{Q\}}\,HP$$

Get

$$(1)\ \Gamma \vdash \ \{x(\mathcal{P}) = x(L)\} \left[\begin{array}{lcl} p \in \mathcal{P} & , & C(p) \\ l \in \{\epsilon\} & , & l \end{array} \right] \left\{ \begin{array}{c} \forall l \in L, \exists p \in \mathcal{P} \\ x(p) = x(l) \end{array} \right\}$$

$$(2)\ \Gamma \vdash \{x(\mathcal{P}) = x(L)\} \left[\begin{array}{lcl} p \in \mathcal{P} & , & C(p) \\ l \in \{x = x + 1; S_1\} & , & l \end{array} \right] \left\{ \begin{array}{c} \forall l \in L, \exists p \in \mathcal{P} \\ x(p) = x(l) \end{array} \right\}$$

where $\Gamma := \{P\}\,[C(\mathcal{P}); S_1]\,\{Q\}$
Apply Weaken-rule to (2):

$$\{x(\mathcal{P}) = x(L)\} \left[\begin{array}{lcl} p \in \{b \mid b \geq 1\} & , & C(p) \\ l \in \{x = x + 1; S_1\} & , & l \end{array} \right] \left\{ \begin{array}{c} \forall l \in L, \exists p \in \mathcal{P} \\ x(p) = x(l) \end{array} \right\}$$

Apply Induction-rule:

$$\frac{xxx}{xxx} Induction$$

1.5 context-free

Given a Doppler attack language L:

$$\begin{array}{rcl} L & \rightarrow & S_1 \\ S_1 & \rightarrow & x = x + 1; S_1; y = y + 1 \\ & & \mid \epsilon \end{array}$$

or in the form of $L = \{(x = x + 1)^n; (y = y + 1)^n \mid n \geq 0\}$
and vulnerable program C:

```
choose n
a=n; b=n
while a>0
    a--
    x = x+1
while b>0
    b--
    y = y+1
```

Write the triple for C:

$$\Gamma \vdash \{x(\mathcal{P}) = x(L) \wedge y(\mathcal{P}) = y(L)\} \left[\begin{array}{ll} p \in \mathcal{P} & , C(p) \\ l \in L & , l \end{array} \right] \left\{ \begin{array}{c} \forall l \in L, \exists p \in \mathcal{P} \\ x(p) = x(l) \wedge y(p) = y(l) \end{array} \right\}$$

where $\mathcal{P} := \mathbb{N}$ and $\Gamma := \{a = b\}$.